F I Z I K A

10-sinf

1. Masala

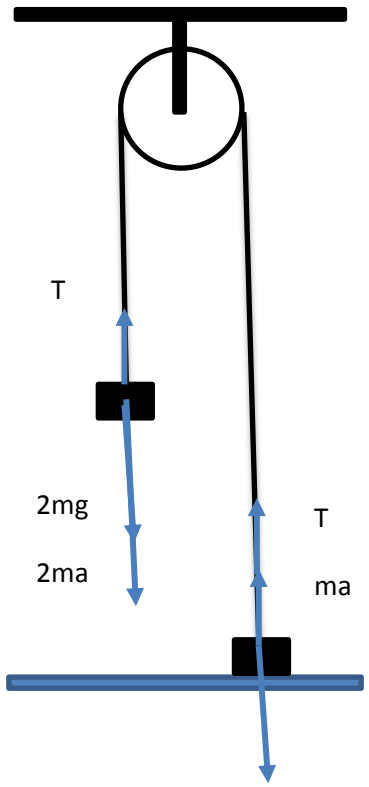
Berilgan:

P

2p

H

h_2 =?



P=mg

2p=2mg bo'ladi.

$$ma = T - mg$$

$$2ma = 2mg - T$$

Bu ikki tenglamalarni qo'shib yuborsak

$$3ma = mg$$

$$a = \frac{g}{3}$$

$$h = \frac{v^2}{2a}$$

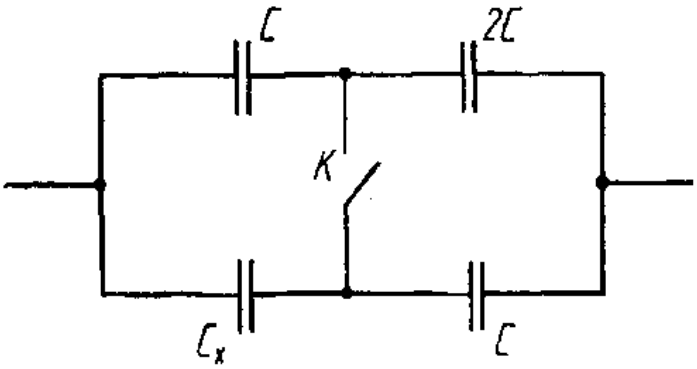
bundan v^2=2ah=$\frac{2gh}{3}$

$$h_x = \frac{v^2}{2g} = \frac{\frac{2gh}{3}}{2g} = \frac{h}{3}$$

$$h_{um} = h + \frac{h}{3} = \frac{4h}{3}$$

Javob: $h_{um} = \frac{4h}{3}$

2. Masala



1-Kalit ulanmagan holda $c_{um} = c_1 + c_2$

$$c_1 = \frac{c * 2c}{c + 2c} = \frac{2c}{3}$$

$$c_2 = \frac{c * c_x}{c + c_x}$$

$$c_{um} = c_1 + c_2 = \frac{2c}{3} + \frac{c * c_x}{c + c_x} = \frac{2c^2 + 5cc_x}{3(c + c_x)}$$

2-Holda esa

$$c_{um} = \frac{c_3 * c_4}{c_3 + c_4}$$

$$c_3 = c + c_x$$
$$c_4 = 3c$$

$$c_{um} = \frac{c_3 * c_4}{c_3 + c_4} = \frac{(c + c_x) * 3c}{4c + c_x}$$

$ikkala\ holatda\ ham\ sig'im\ o'zmaganligi\ uchun$
$\frac{2c^2+5cc_x}{3(c+c_x)} = \frac{(c+c_x)*3c}{4c+c_x}$

$$(2c^2 + 5cc_x) * (4c + c_x) = (c + c_x)^2 * 9c$$

Bu tenglama qavslarini ochib chiqib soddalashtirsak

$$c^3 - 4c^2c_x + 4cc_x{}^2 = 0$$
$$c^2 - 4cc_x + c_x{}^2 = 0$$
$$(c - 2c_x)^2 = 0$$
$$c_x = \frac{c}{2}$$

Javob:

$$c_x = \frac{c}{2}$$

11-sinf

1. Masala

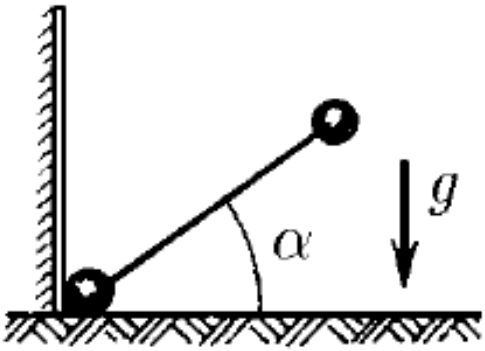
Berilgan:

m

g

α

F=?



$$ma = T + mgsin\alpha$$

$$h = l * sin\alpha$$

$$mgl = \frac{m * v^2}{2} + mglsin\alpha$$

$$v^2 = 2gl(1 - sin\alpha)$$

$$a = \frac{v^2}{l}$$

$$T = \frac{mv^2}{l} - mgsin\alpha$$

$$T = mg(2 - 3sin\alpha)$$

$$F = Tcos\alpha$$

$$F = mgcos\alpha\ |3\sin\alpha - 2|$$

Agar $\sin\alpha \geq \frac{2}{3}$ bo'lsa F=0 bo'ladi

Javob:

$$F = mgcos\alpha\ |3\sin\alpha - 2|$$

Agar $\sin\alpha \geq \frac{2}{3}$ bo'lsa F=0

2. Masala

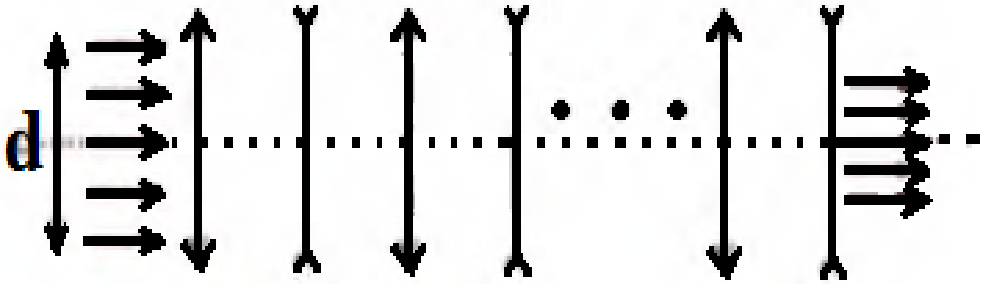
Berilgan:

$\boldsymbol{F_1} = 9$ sm

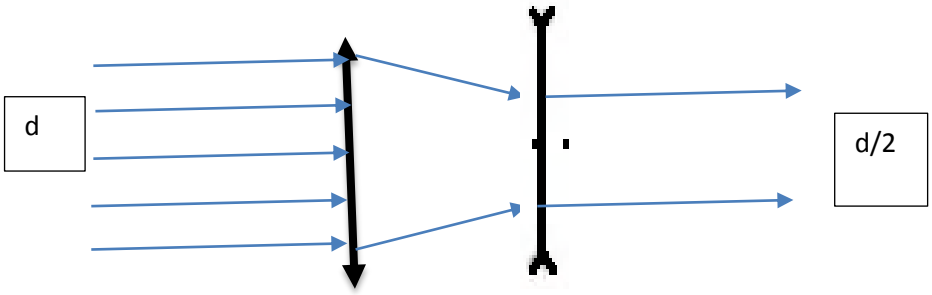
$\boldsymbol{F_2} = 4{,}5$ sm

a= 4,5 sm

d= 160 sm



Birinchi yig'uvchi va ikkinchi linza sochuvchi linza bo'lganligi va yig'uvchi linzaning fokus masofasi sochuvchi linzanikidan ikki marta katta bo'lganligi uchun. Va orasidagi masofa sochuvchi linza fokusiga tengligini hisobga olsak quyidagicha chizma bo'ladi natijaviy optic kuch esa nolga teng bo'lib qoladi. Bu esa parallel kelgan nurlar yana parallel ketishini bildiradi.



Yig'uvchi linzada fokus masofada nur yig'iladi fokusning yarmida esa nurning diametric ham ikki marta kayadi. Shuning uchun har bir juft yig'uvchi va sochuvchi linza sistemasida nur diametric 2 marta kamayadi. Butun sistemada esa 5 ta juft linzalar borligi sababli

$$d_n = \frac{d}{2^5} = \frac{160}{32} = 5\ sm$$

Javob: $d_n = 5\ sm$

@olimpdep
Fan olimpiadalari bo'yicha iqtidorli o'quvchilar bilan ishlash DEPARTAMENTI